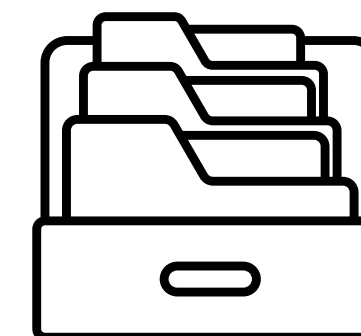
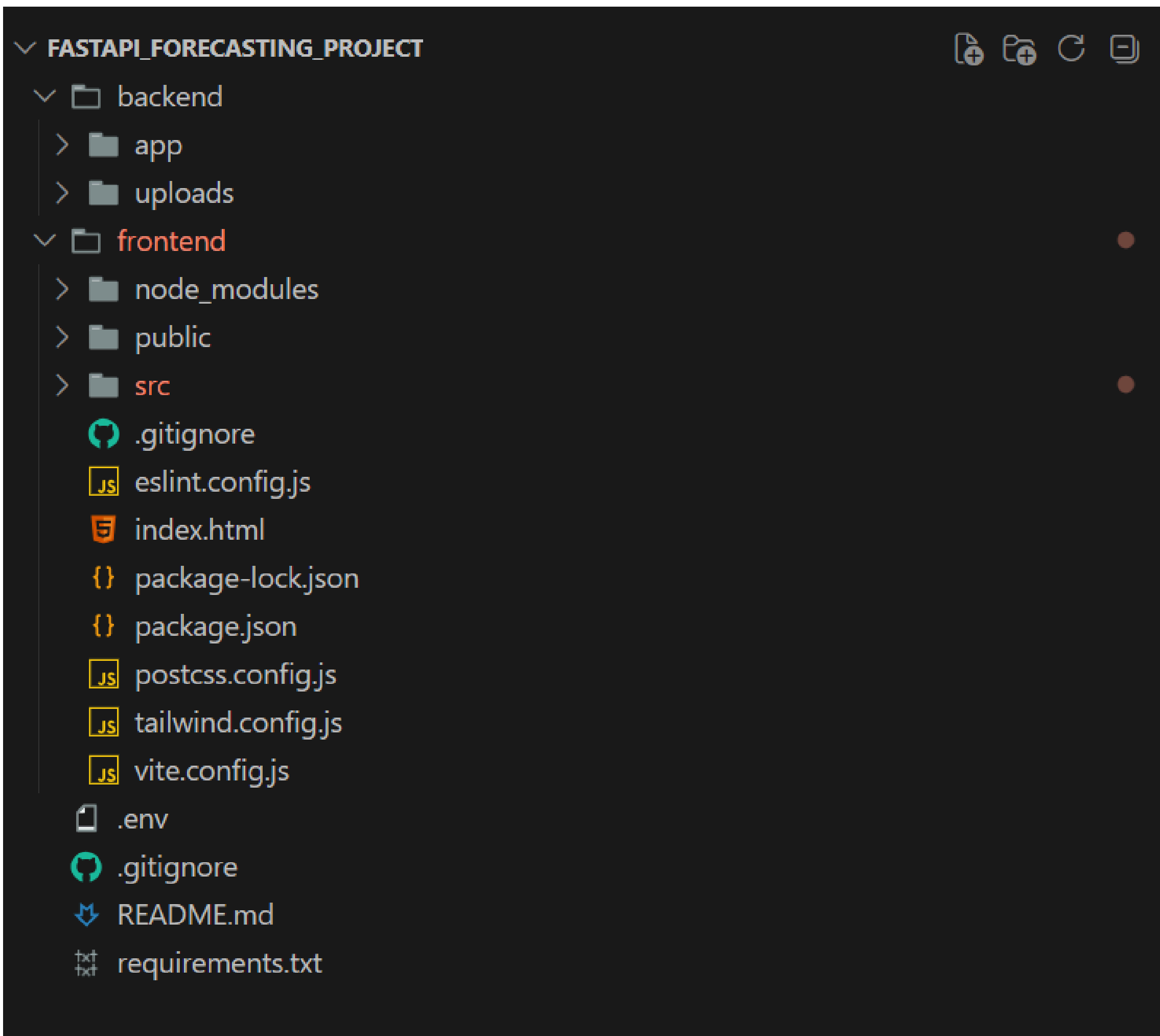
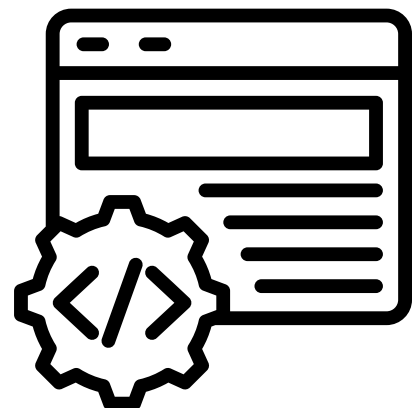




ARCHITECTURE



ENDPOINT DIVISION

AI Driven Demand Forecasting & Prediction Web App 0.1.0 OAS 3.1

/openapi.json

Authorize



auth

POST /auth/register Register

POST /auth/login Login

datasets

POST /datasets/upload Upload Dataset

forecast

GET /forecast/{product_name} Get Forecast

analytics

GET /analytics/summary Get Summary

reports

GET /reports/export/excel Export Excel

default

GET / Read Root

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/auth/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "jerry",
    "email": "jerry@abc.com",
    "password": "abc"
  }'
```

Request URL

```
http://127.0.0.1:8000/auth/register
```

Server response

Code	Details
------	---------

200

Response body

```
{
  "id": 5,
  "username": "jerry",
  "email": "jerry@abc.com",
  "is_active": true
}
```



Download

Response headers

```
access-control-allow-credentials: true
content-length: 68
content-type: application/json
date: Thu, 14 May 2026 19:33:23 GMT
server: uvicorn
```

Responses

SIGN UP / REGISTER

SIGN IN

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/auth/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=jerry%40abc.com&password=abc&scope=&client_id=string&client_secret=*****'
```

Request URL

http://127.0.0.1:8000/auth/login

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqZXJyeUBhYmMuY29tIiwiaXNjaXNzIjoxNzc4Nzg5MDU4fQ.k1qVNzN5auCG1oPiHTix8FQuANhcutxJG6YUQTHrwcE",
  "token_type": "bearer"
}
```



Download

Response headers

```
access-control-allow-credentials: true
content-length: 176
content-type: application/json
date: Thu, 14 May 2026 19:34:17 GMT
server: uvicorn
```

Responses

Code	Description
------	-------------

200	Successful Response
-----	---------------------

Links

4

No links

$\wedge$

Reset

multipart/form-data

 cafe_sales.csv

Clear

```
curl -X 'POST' \
'http://127.0.0.1:8000/datasets/upload' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJyYWtlczhaZ21haWwvY29tIiwiaWF0IjoxNzc4Nzg4NzA5fQ.GABj7RI LSsRFuW40TK7hdZENhCILUIwSmJUJbrw0-j4' \
-H 'Content-Type: multipart/form-data' \
-F 'file=@cafe_sales.csv;type=text/csv'
```

<http://127.0.0.1:8080/datasets/upload>

Code Details

```
{
  "status": "success",
  "rows_uploaded": 16,
  "file_saved_at": "app/uploads\\cafe_sales.csv"
}
```

 Download

```
access-control-allow-credentials: true
content-length: 85
content-type: application/json
date: Thu, 14 May 2026 19:29:02 GMT
server: uvicorn
```

Links

5

GET /forecast/{product_name} Get Forecast

Parameters

Cancel

Name	Description
product_name * required	
string (path)	Masala Chai Latte
days	
integer (query)	11 maximum: 30

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/forecast/Masala%20Chai%20Latte?days=11' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/forecast/Masala%20Chai%20Latte?days=11
```

Server response

Code	Details
200	Response body <pre>{ "product": "Masala Chai Latte", "historical_count": 4, "prediction_days": 11, "next_forecast": [20.5, 16.2, 11.9, 7.6, 3.3, 0, 0, 0, 0, 0, 0] }</pre> Response headers

Download

LINEAR REGRESSION FORECAST FOR 'X' DAYS

ANALYTICS/ PRODUCT SUMMARY

analytics

GET

/analytics/summary

Get Summary

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/analytics/summary' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJyYWtlczhAZ21hbmVudY29tIiwiaXNjaXNzIjoxNzc4Nzg4NzA5fQ.GABj7RIL5sRfUw48TK7hdZEMhCILIwSnJUJbrw0-j4'
```

Request URL

http://127.0.0.1:8000/analytics/summary

Server response

Code	Details
200	Response body <pre>{ "total_revenue": 73450, "total_sales": 399, "top_products": [{ "name": "Masala Chai Latte", "value": 125 }, { "name": "Paneer Tikka Tacos", "value": 115 }, { "name": "Za'atar Grilled Chicken", "value": 109 }] }</pre> Download Response headers <pre>content-length: 190 content-type: application/json date: Thu, 14 May 2026 19:31:14 GMT server: uvicorn</pre>

Responses



forecast_report (3).xlsx

5.7 KB • Done

GET /reports/export/excel Export Excel

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/reports/export/excel' \
  -H 'accept: application/json'
```



Request URL

http://127.0.0.1:8000/reports/export/excel

Server response

Code Details

200

Response body

[Download file](#)

Response headers

```
content-disposition: attachment; filename=forecast_report.xlsx
content-type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
date: Thu, 14 May 2026 19:31:52 GMT
server: uvicorn
transfer-encoding: chunked
```

Responses

Code Description

Links

200

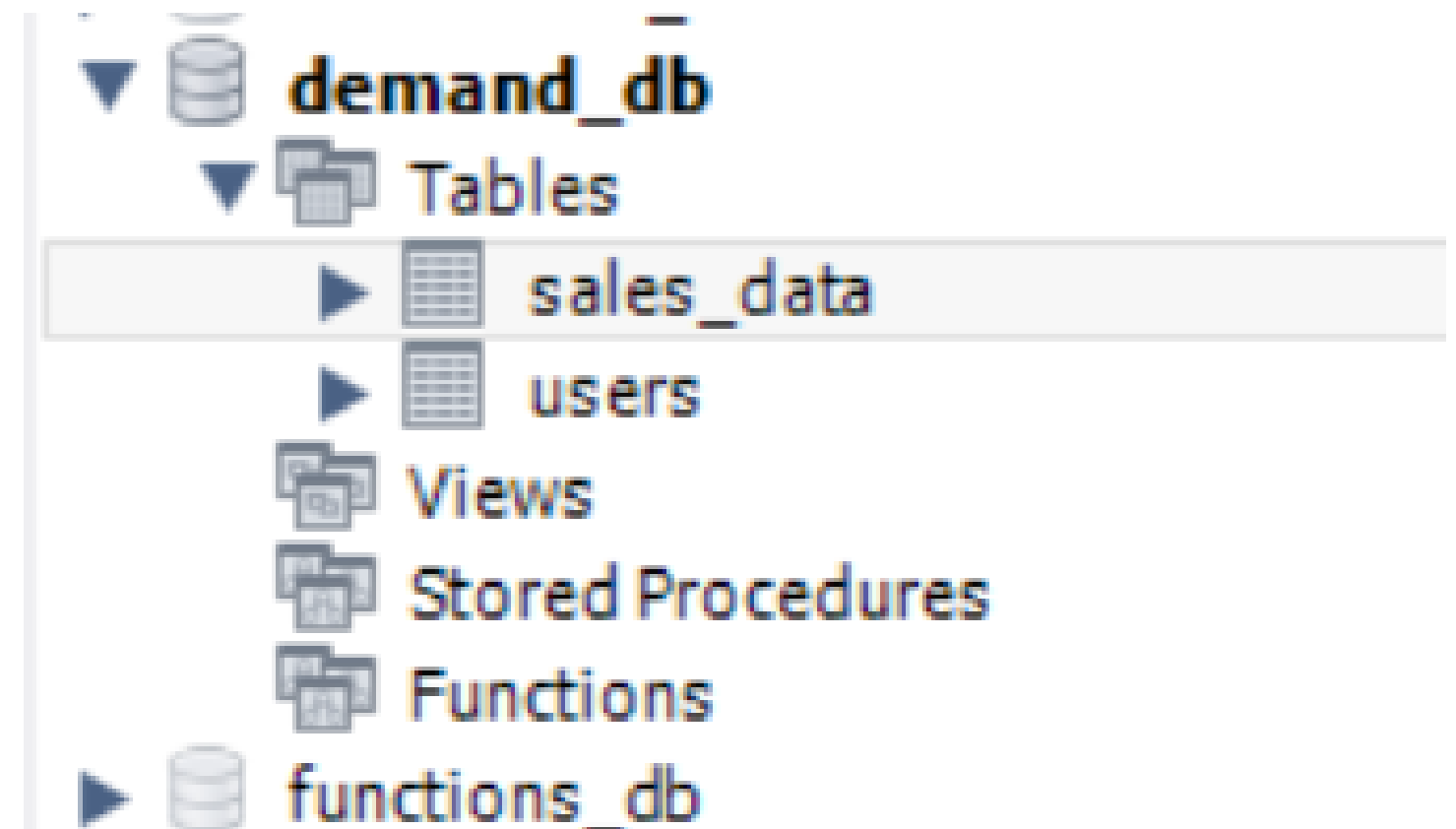
Successful Response

No links

Media type

REPORT EXPORT

MYSQL INTEGRATION



```
● PS C:\Users\Hp\OneDrive\Desktop\FastAPI_Forecasting_Project> cd backend
● PS C:\Users\Hp\OneDrive\Desktop\FastAPI_Forecasting_Project\backend> cd app
● PS C:\Users\Hp\OneDrive\Desktop\FastAPI_Forecasting_Project\backend\app> python test_db.py
Connecting to: mysql+pymysql://root:SSabMySQL%407860@localhost:3306/demand_db
✓ SUCCESS: MySQL is connected to your project!
❖ PS C:\Users\Hp\OneDrive\Desktop\FastAPI_Forecasting_Project\backend\app>
```

Query 1

SQL File 3*

SQL File 5*

SQL File 5*

SQL File 6*

task4

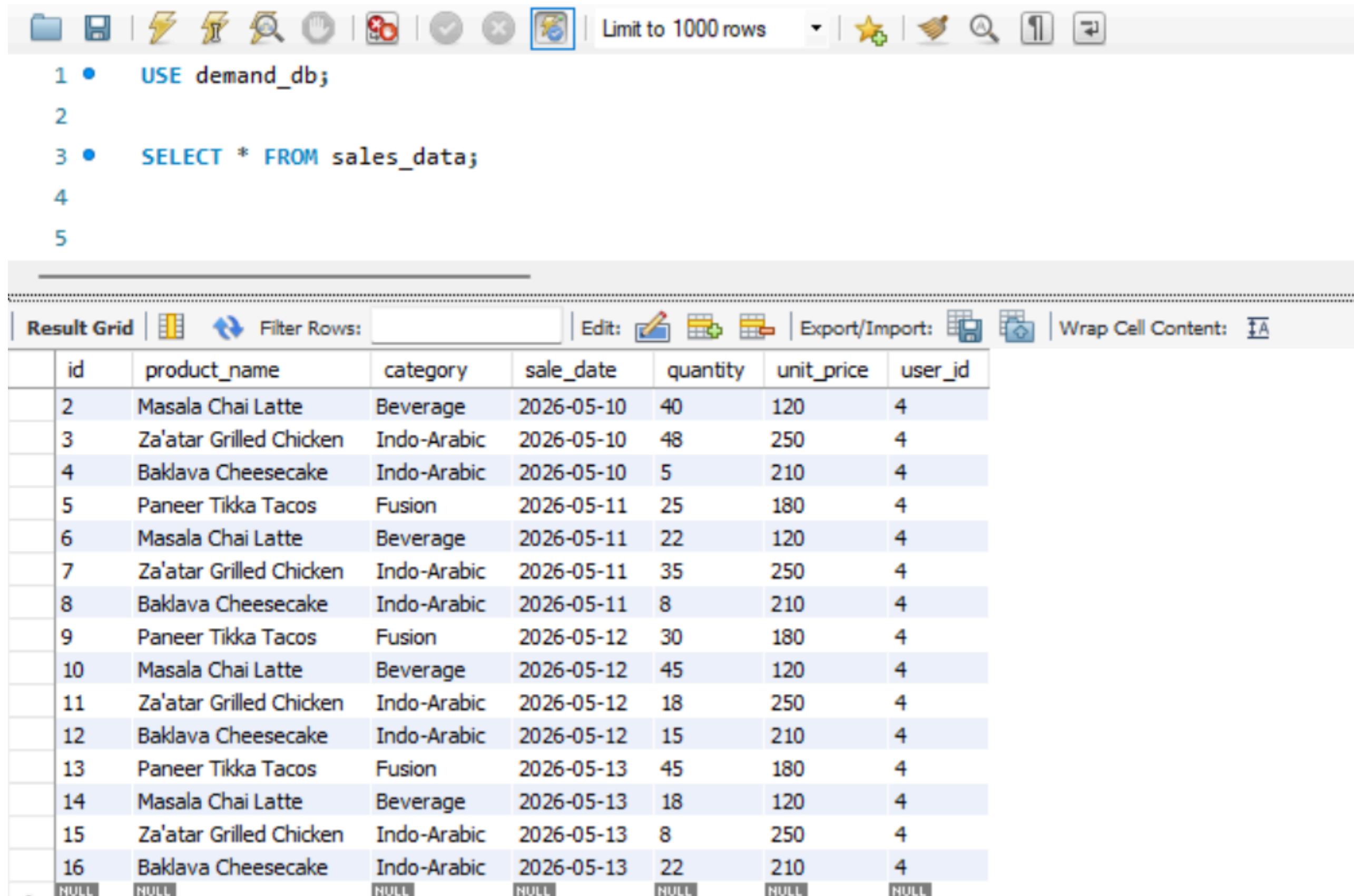
task5

task6

</

MYSQL USERS TABLE

MYSQL SALES_DATA TABLE



The screenshot shows a MySQL IDE interface. At the top, there is a toolbar with various icons for file operations, execution, and navigation. Below the toolbar, a SQL query is entered in a text area:

```
1 • USE demand_db;
2
3 • SELECT * FROM sales_data;
4
5
```

Below the query editor, the 'Result Grid' tab is active, displaying the results of the query. The results are shown in a table with 8 columns: id, product_name, category, sale_date, quantity, unit_price, and user_id. The table contains 16 rows of data, followed by a row of NULL values. The table is styled with alternating light blue and white rows.

	id	product_name	category	sale_date	quantity	unit_price	user_id
	2	Masala Chai Latte	Beverage	2026-05-10	40	120	4
	3	Za'atar Grilled Chicken	Indo-Arabic	2026-05-10	48	250	4
	4	Baklava Cheesecake	Indo-Arabic	2026-05-10	5	210	4
	5	Paneer Tikka Tacos	Fusion	2026-05-11	25	180	4
	6	Masala Chai Latte	Beverage	2026-05-11	22	120	4
	7	Za'atar Grilled Chicken	Indo-Arabic	2026-05-11	35	250	4
	8	Baklava Cheesecake	Indo-Arabic	2026-05-11	8	210	4
	9	Paneer Tikka Tacos	Fusion	2026-05-12	30	180	4
	10	Masala Chai Latte	Beverage	2026-05-12	45	120	4
	11	Za'atar Grilled Chicken	Indo-Arabic	2026-05-12	18	250	4
	12	Baklava Cheesecake	Indo-Arabic	2026-05-12	15	210	4
	13	Paneer Tikka Tacos	Fusion	2026-05-13	45	180	4
	14	Masala Chai Latte	Beverage	2026-05-13	18	120	4
	15	Za'atar Grilled Chicken	Indo-Arabic	2026-05-13	8	250	4
	16	Baklava Cheesecake	Indo-Arabic	2026-05-13	22	210	4
	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Forecastly.

Precision in Every Byte.

Redefining AI-driven demand forecasting.



System Active
Ready for data ingestion

Welcome Back...

Please enter your credentials

abdul99@gmail.com



.....|



[Terms & Conditions](#)

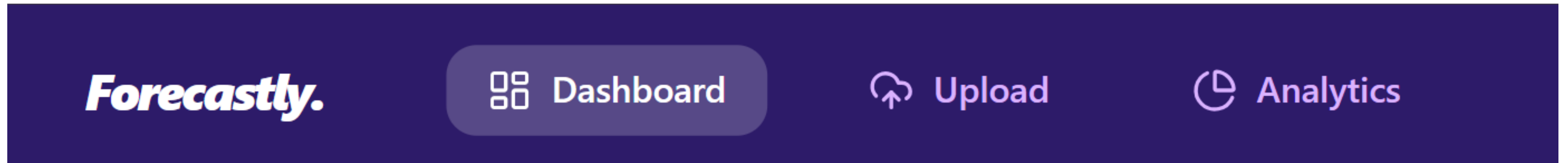
[Forgot Password?](#)

login... →

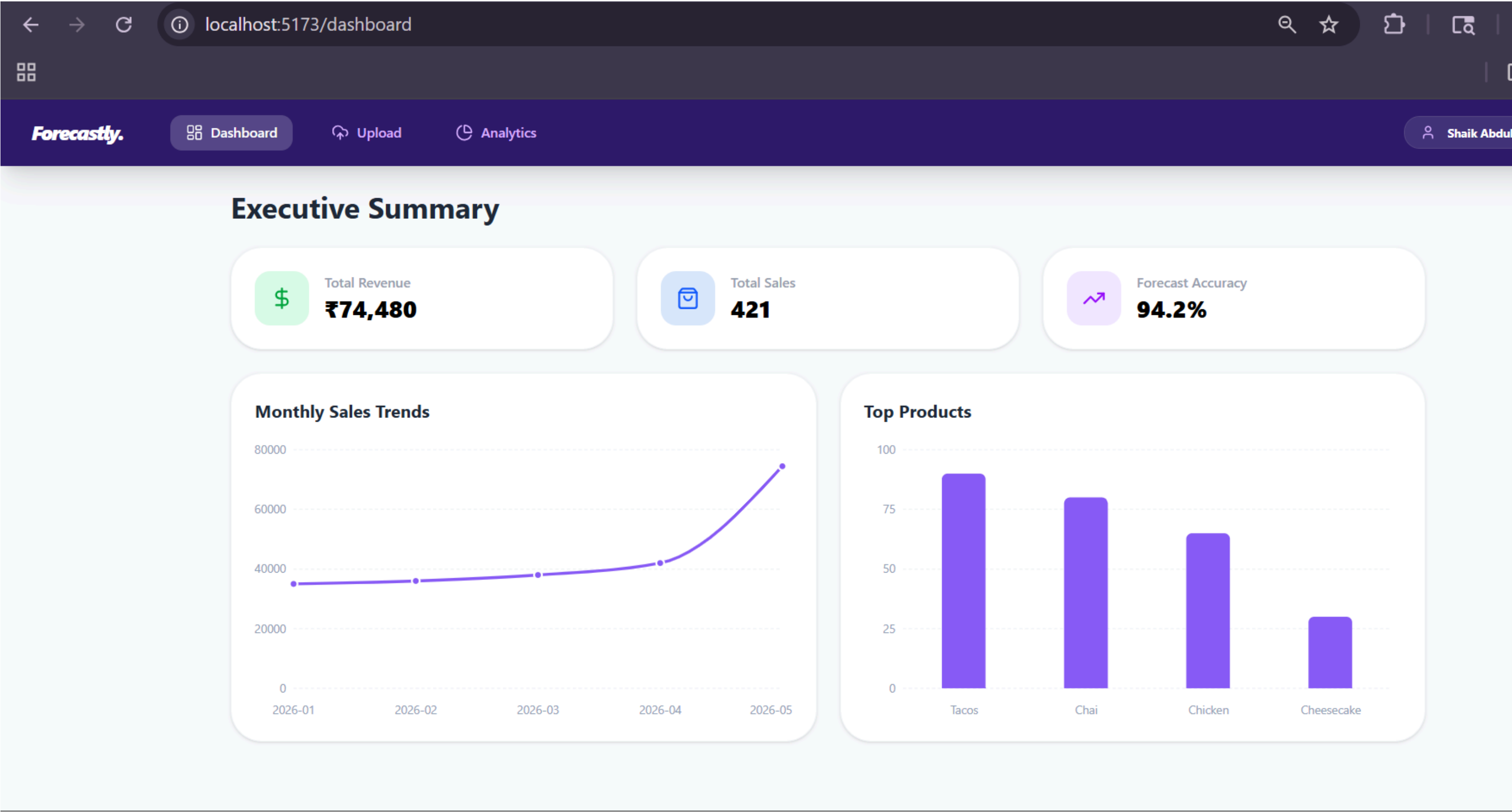
New here? [Create Account](#)

FORECASTLY- LOGIN PAGE

HORIZONTAL NAV BAR



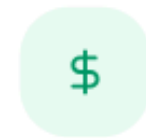
DASHBOARD VAR1



Executive Summary

Real-time performance metrics for your outlet.

⚡ AI Retraining Active



TOTAL REVENUE

₹73,450



TOTAL SALES

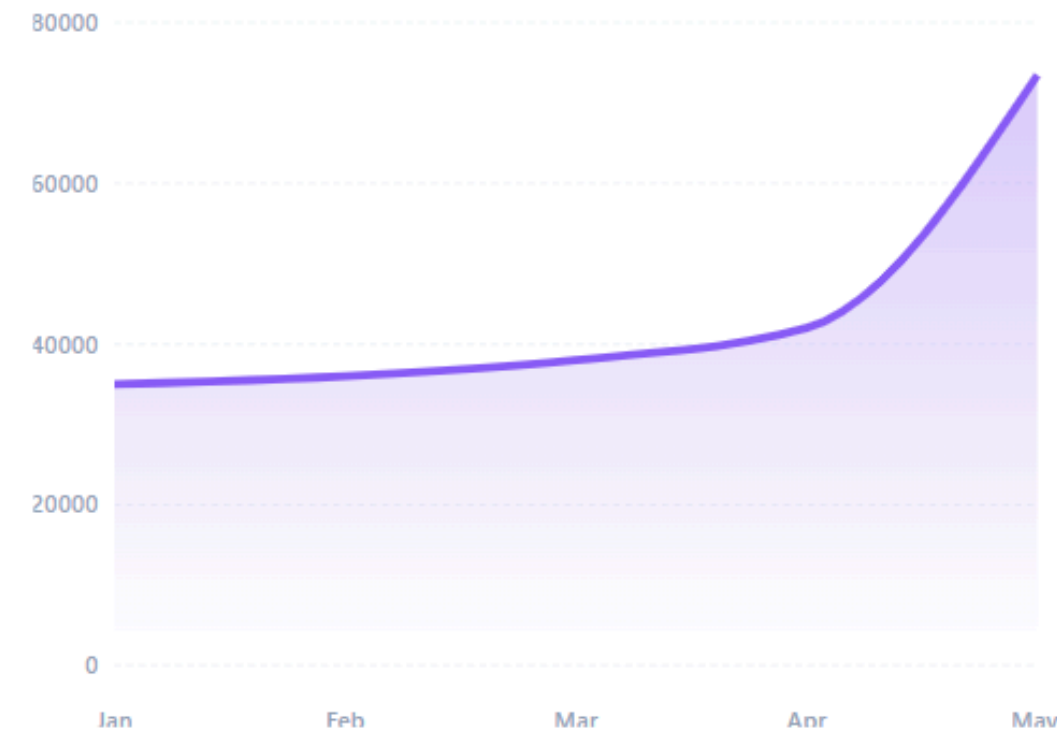
399



FORECAST ACCURACY

94.2%

Revenue Momentum



Signature Performance

RANK #1

Paneer Tikka Tacos

150 UNITS

RANK #2

Masala Chai Latte

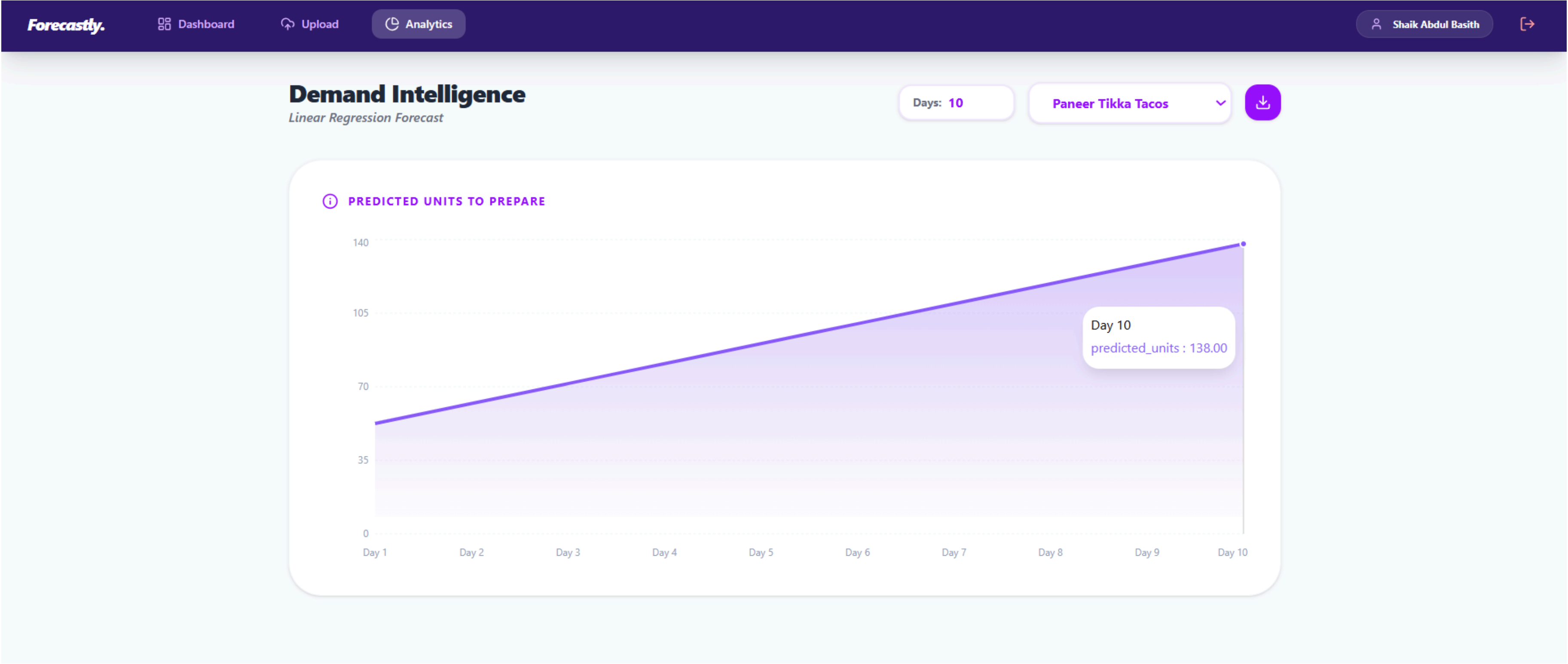
120 UNITS

RANK #3

Za'atar Grilled Chicken

85 UNITS

DASHBOARD VAR2



PANEER 10 DAY FORECAST - WINNER

TEA 7 DAY FORECAST

Demand Intelligence

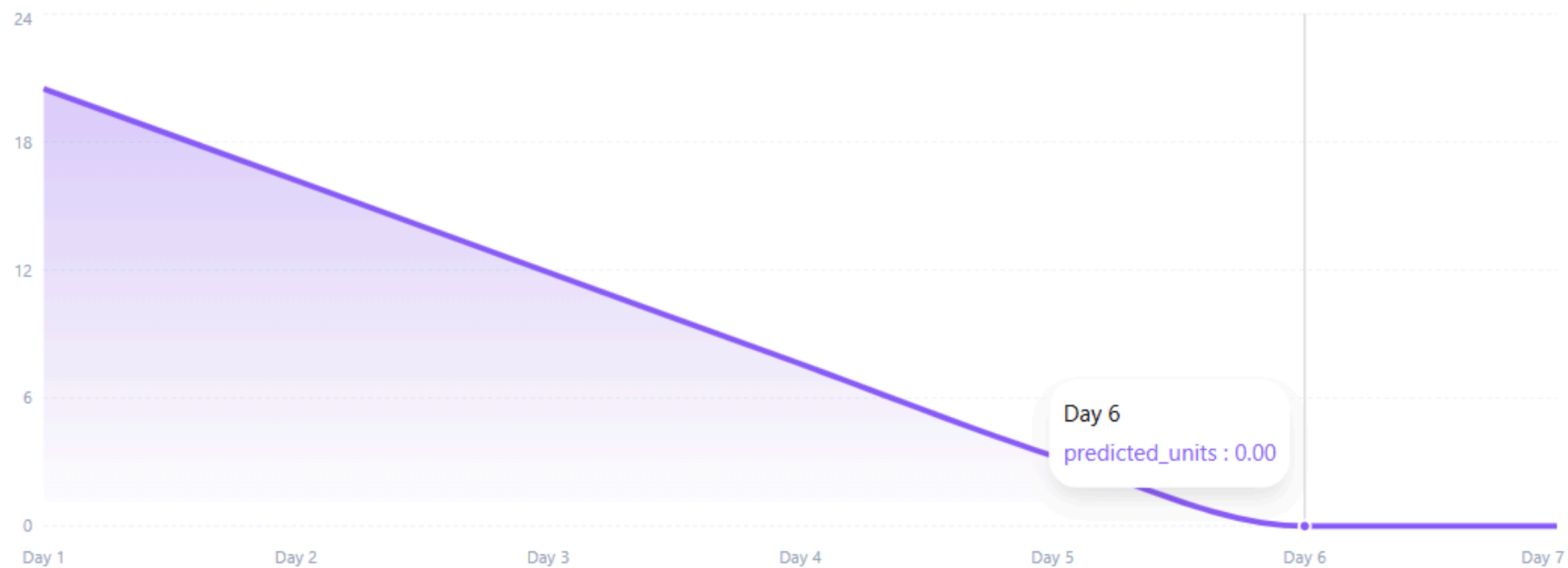
Linear Regression Forecast

Days: 7

Masala Chai Latte



PREDICTED UNITS TO PREPARE



Demand Intelligence

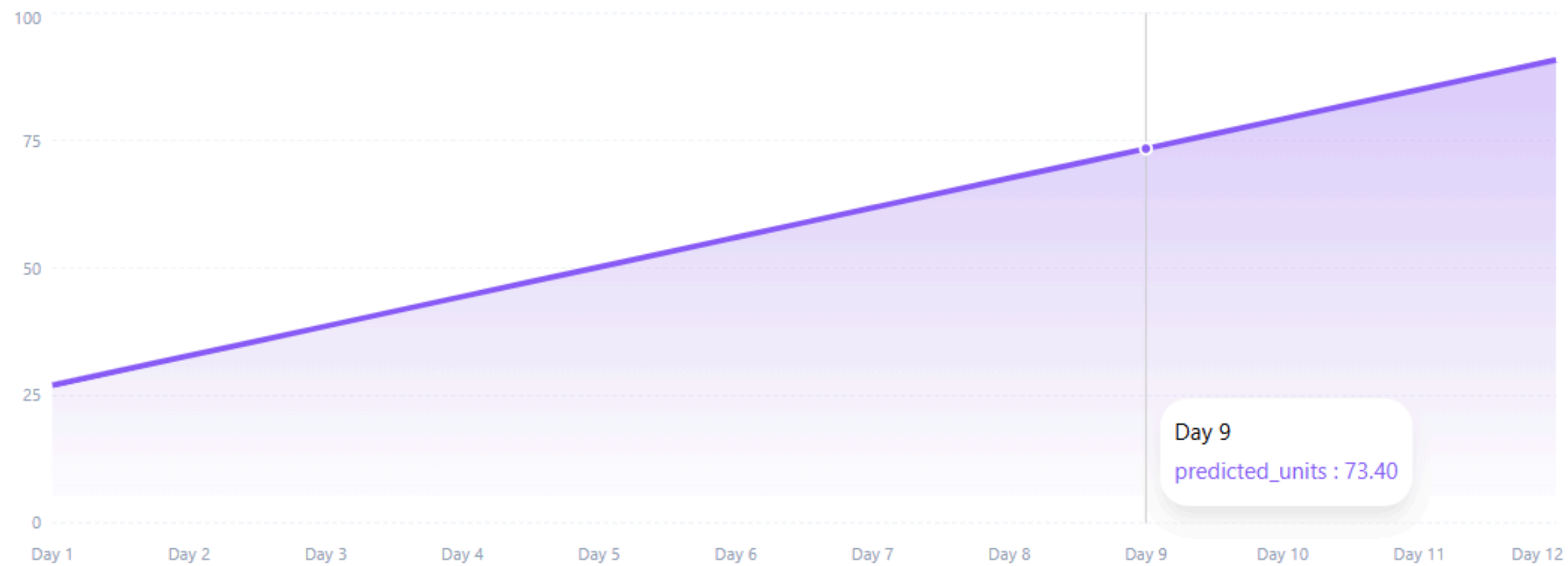
Linear Regression Forecast

Days: 12

Baklava Cheesecake

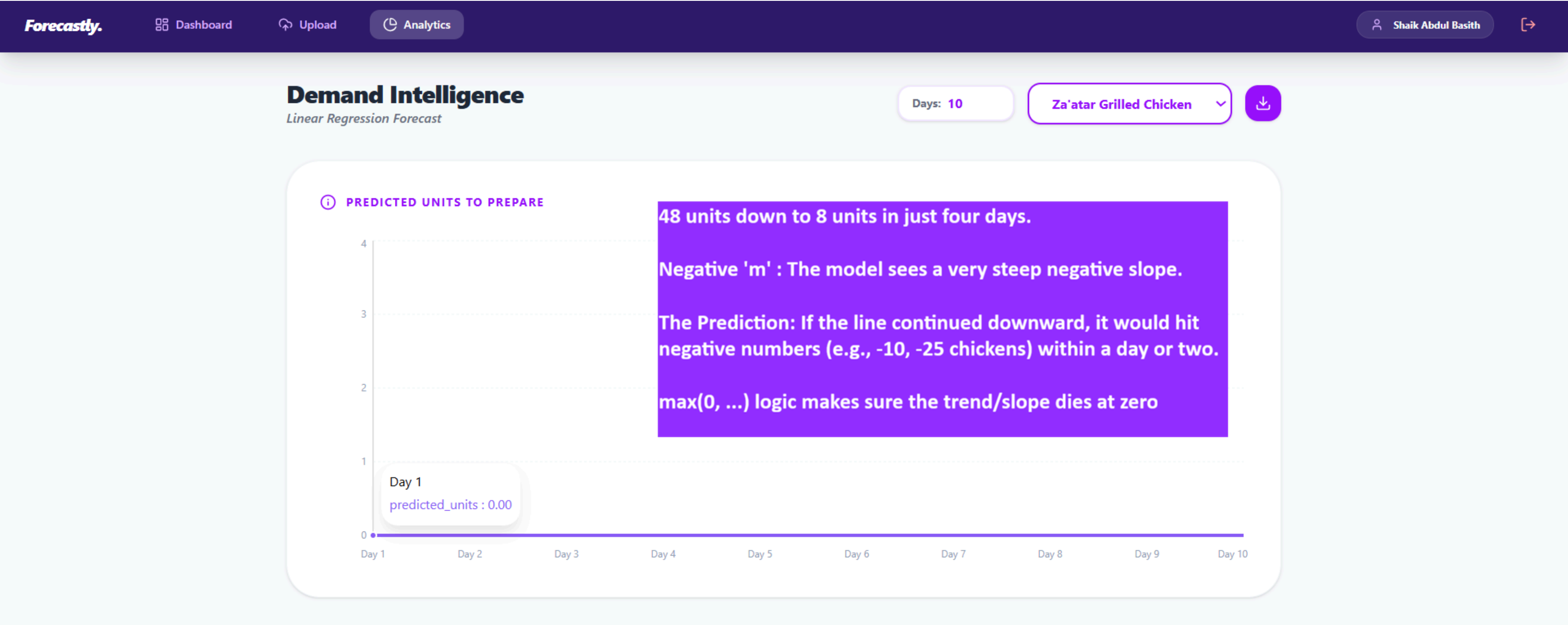


PREDICTED UNITS TO PREPARE

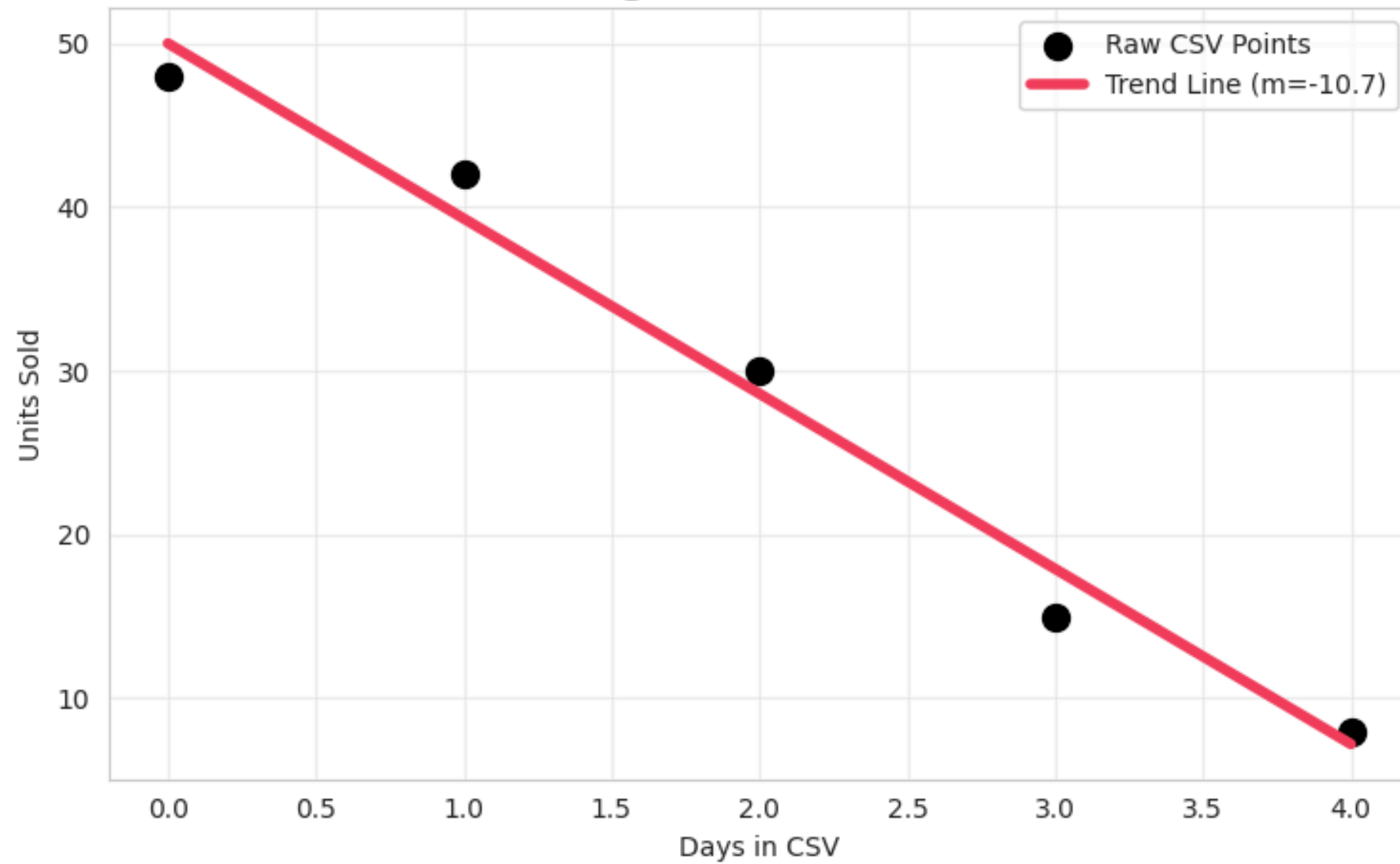


CHEESE CAKE 12 DAY FORECAST

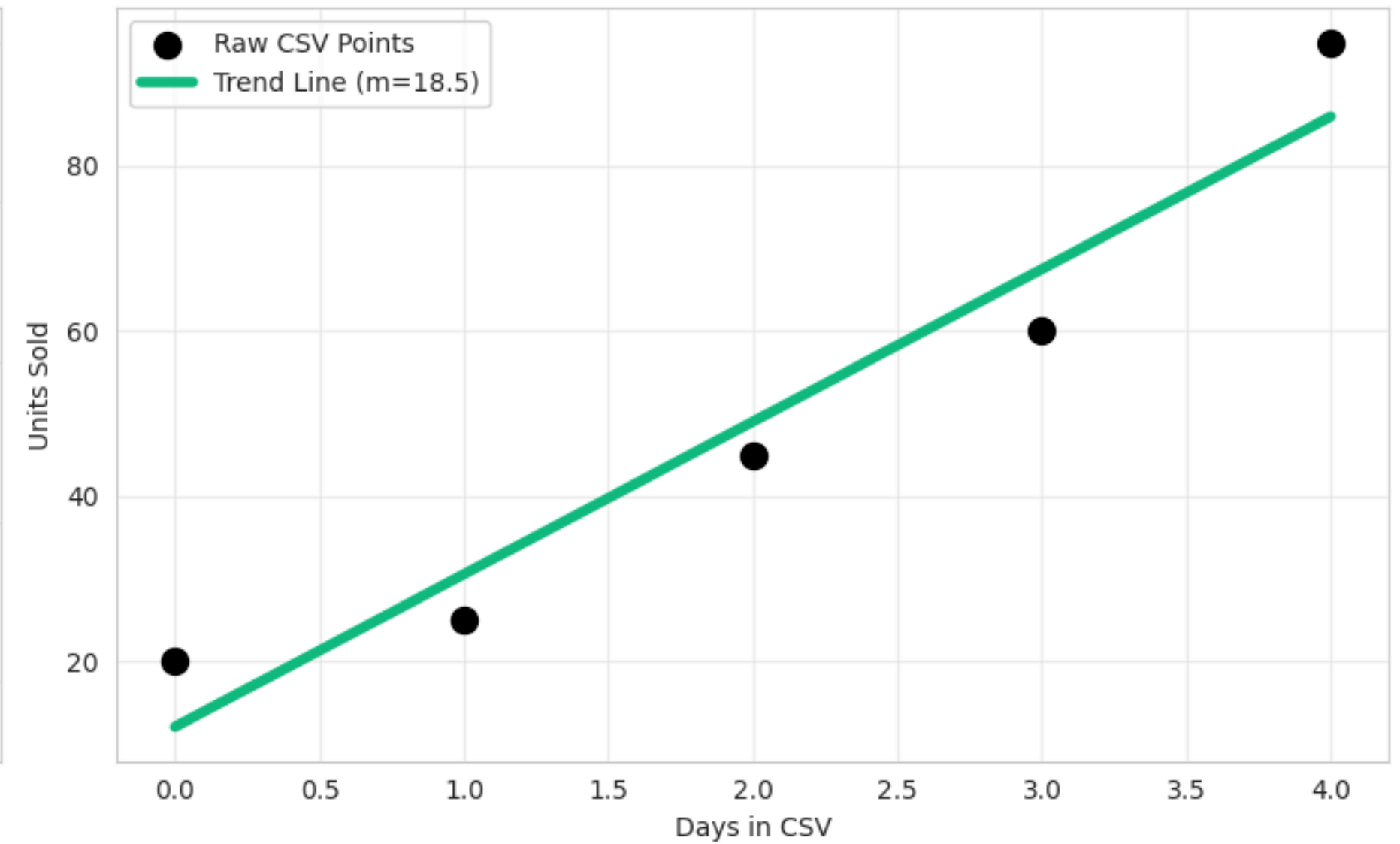
CHICKEN 10 DAY FORECAST - UNSTABLE / LOSER



Proof: Negative Trend (Chicken)



Proof: Positive Trend (Tacos)



LINEAR REGRESSION IN ACTION

REPORT DOWNLOAD FUNCTIONALITY

